Data_Snooping_AC

Helen Payne

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```r
# Load necessary library
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

DATA SNOOPING: looking into survival Next we wanted to add in survival into AC_22_23_fit, but in order to do this, we need to get the data from AC_22_23_full

```r
#read in both datasets
AC_22_23_full <- read_csv(here::here("data_sheets", "compiled_sheets", "AC_22_23_full.csv"))
```

```
## Rows: 6733 Columns: 65
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (7): Gen, Donor, Recipient, sample_ID_SEG, SegPos, Block, Replicated
## dbl  (45): Year, Transect, Sequence, Plant_ID, days_sow2flower, days_plant2f...
## lgl   (4): F_plant, F_Num_03, Rep_FitP, any_FitP
## date  (9): Sow_Date, Plant_Date, FFD, LFD, F_Num_01, F_Num_02, photo_date, p...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
AC_22_23_fit <- read_csv(here::here("data_sheets", "compiled_sheets", "AC_22_23_fit.csv"))
```

```
## Rows: 864 Columns: 65
## -- Column specification --------------------------------------------------------
```

```
## Delimiter: ","
## chr   (6): Gen, Donor, Recipient, sample_ID_SEG, SegPos, Replicated
## dbl  (46): Year, Transect, Sequence, Plant_ID, days_sow2flower, days_plant2f...
## lgl   (4): F_plant, F_Num_03, Rep_FitP, any_FitP
## date  (9): Sow_Date, Plant_Date, FFD, LFD, F_Num_01, F_Num_02, photo_date, p...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

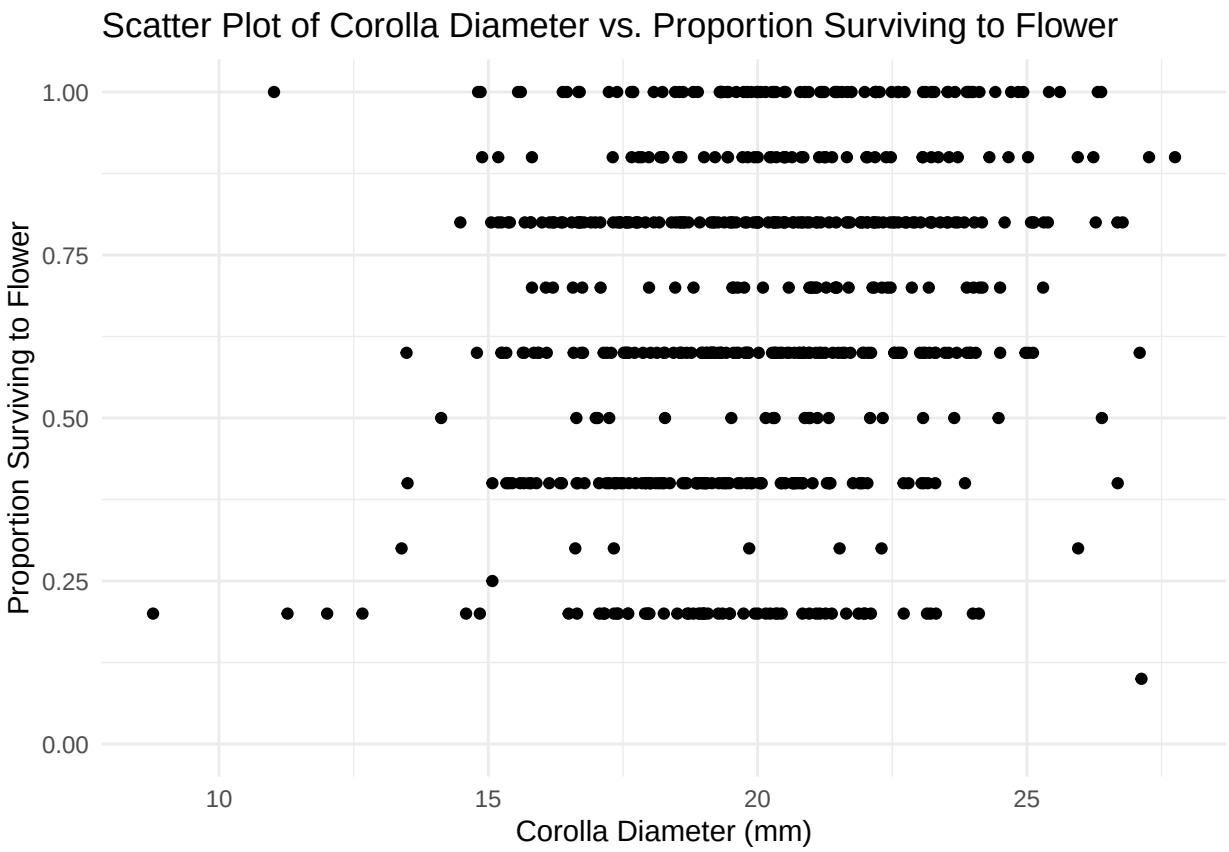
```r
# Calculate the mean surv_to_flower for each group in AC_22_23_full and add it as a new column
AC_22_23_full <- AC_22_23_full %>%
  group_by(Year, Gen, Transect, Sequence, Donor, Recipient, SegPos) %>%
  mutate(prop_surv_to_flower = mean(surv_to_flower, na.rm = TRUE)) %>%
  ungroup()  # Remove grouping

AC_prop_sample <- AC_22_23_full %>%
  select(c(Year, Gen, Transect, Sequence, Donor, Recipient, SegPos, prop_surv_to_flower))  %>%
    distinct()

AC_22_23_fit <- AC_22_23_fit %>%
  left_join(AC_prop_sample %>% select(Year, Gen, Transect, Sequence, Donor, Recipient, SegPos, prop_surv
            by = c("Year", "Gen", "Transect", "Sequence", "Donor", "Recipient", "SegPos"))
```

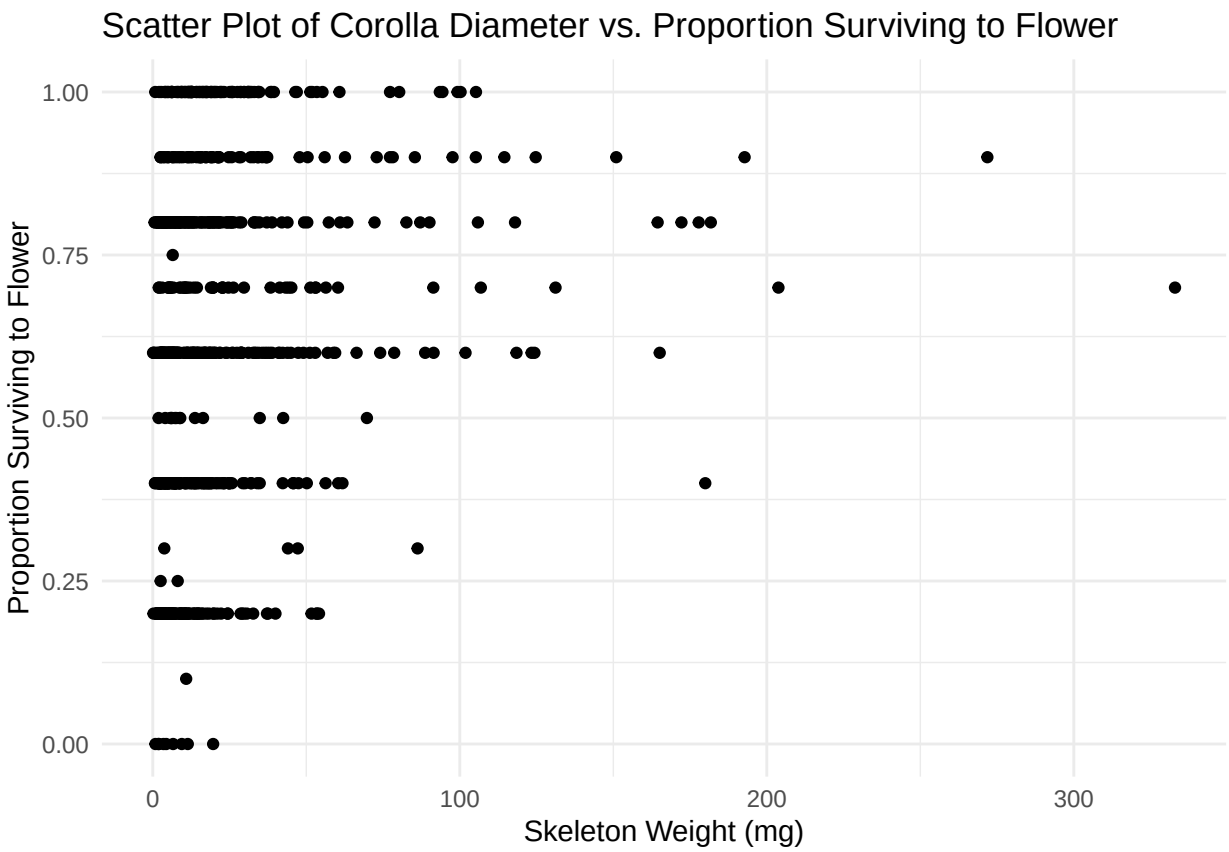
```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = corolla_diam_mm_SEG, y = prop_surv_to_flower)) +
  geom_point() +
  labs(
    title = "Scatter Plot of Corolla Diameter vs. Proportion Surviving to Flower",
    x = "Corolla Diameter (mm)",
    y = "Proportion Surviving to Flower"
  ) +
  theme_minimal()
```

```
## Warning: Removed 269 rows containing missing values (`geom_point()`).
```



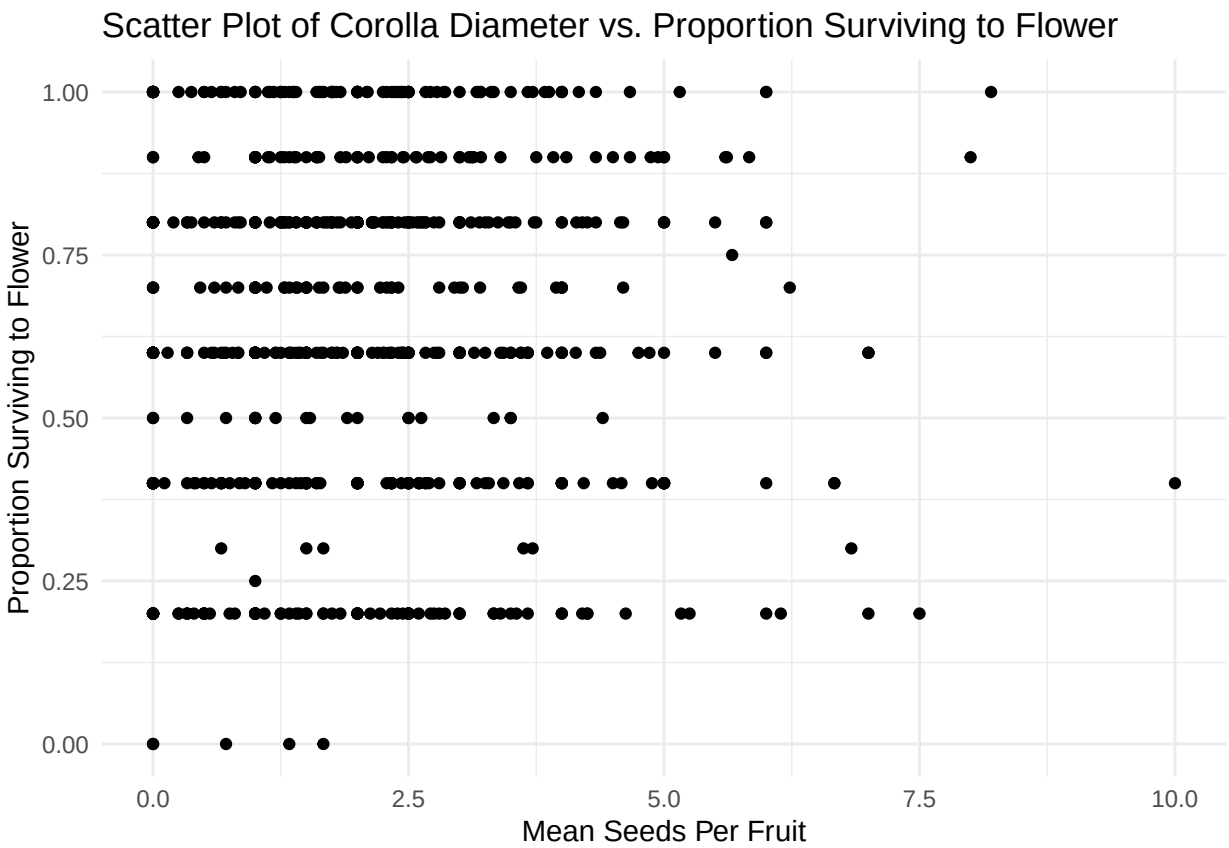

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = skel_dryweight_mg_SEG, y = prop_surv_to_flower)) +
  geom_point() +
  labs(
    title = "Scatter Plot of Corolla Diameter vs. Proportion Surviving to Flower",
    x = "Skeleton Weight (mg)",
    y = "Proportion Surviving to Flower"
  ) +
  theme_minimal()
```

```
## Warning: Removed 163 rows containing missing values (`geom_point()`).
```



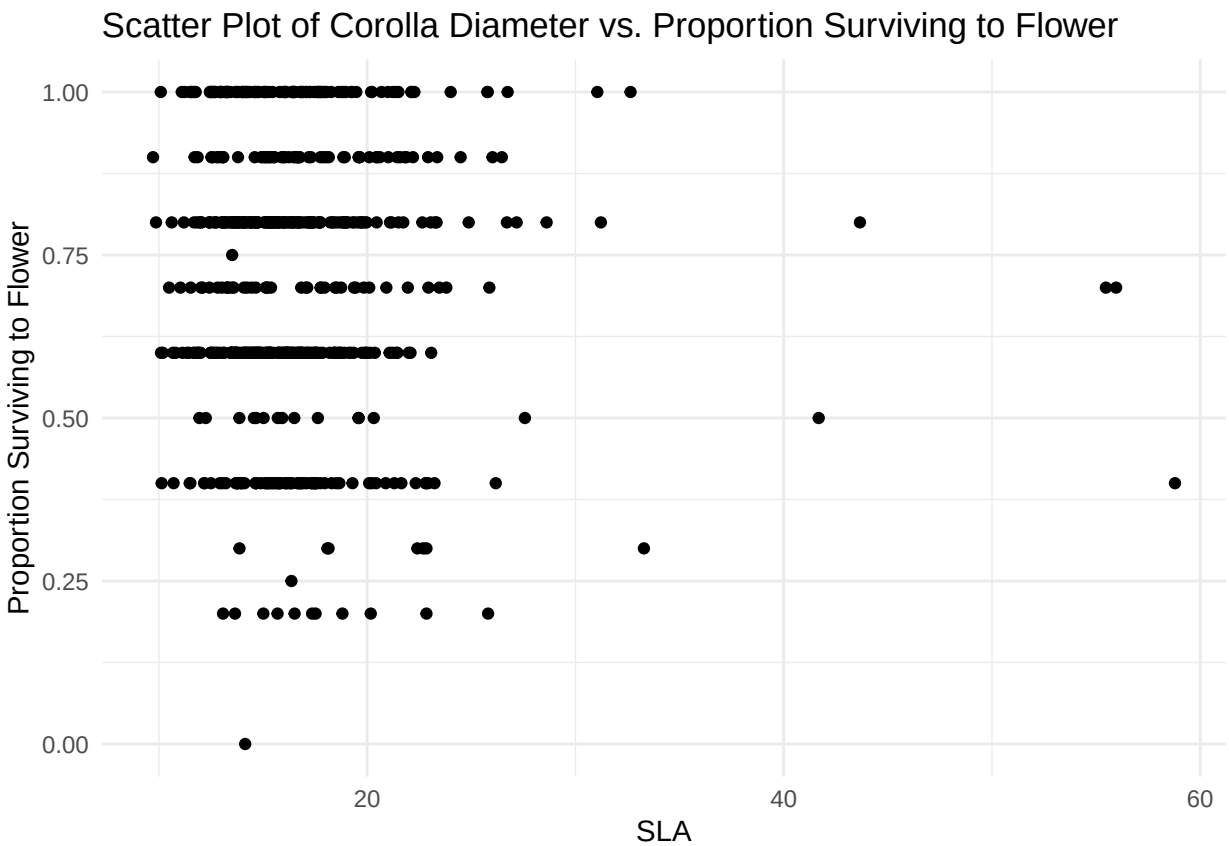

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = mean_seeds_per_fruit, y = prop_surv_to_flower)) +
  geom_point() +
  labs(
    title = "Scatter Plot of Corolla Diameter vs. Proportion Surviving to Flower",
    x = "Mean Seeds Per Fruit",
    y = "Proportion Surviving to Flower"
  ) +
  theme_minimal()
```

```
## Warning: Removed 150 rows containing missing values (`geom_point()`).
```



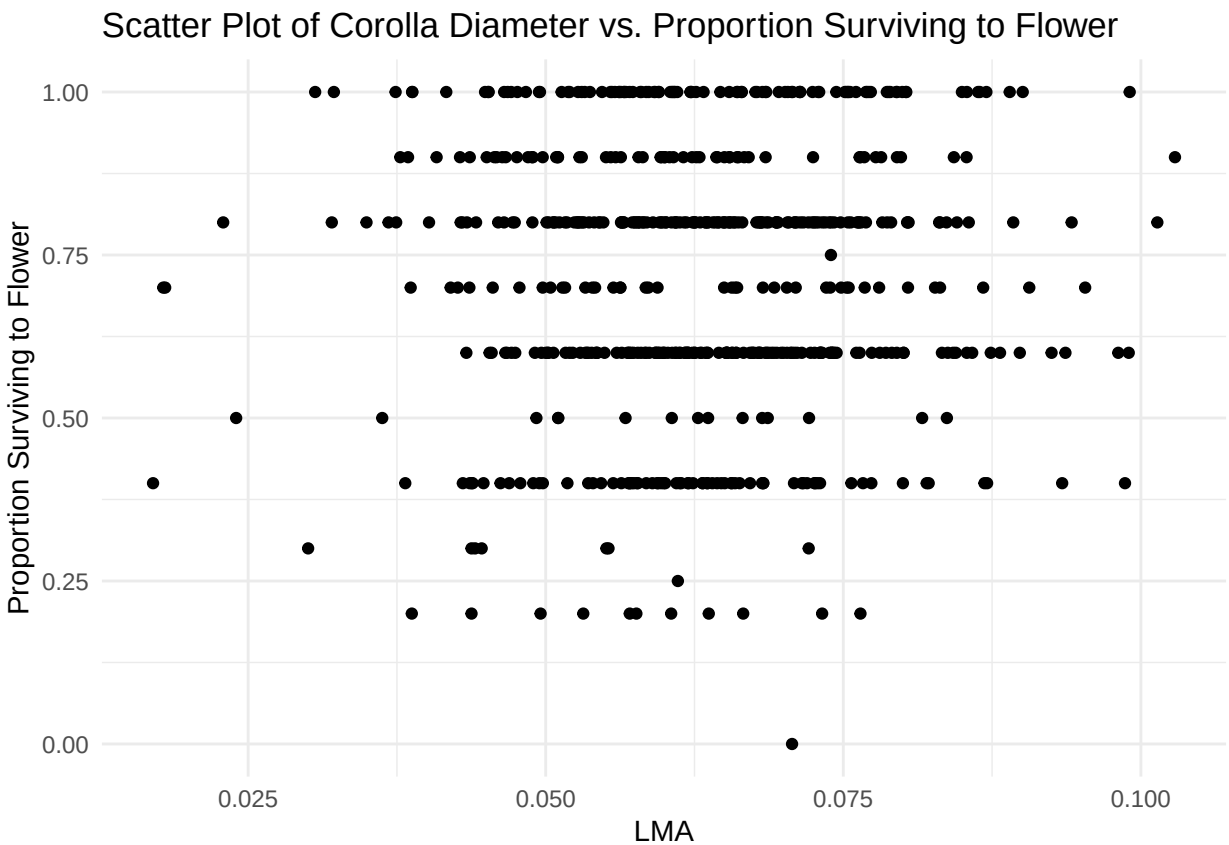

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = SLA_SEG, y = prop_surv_to_flower)) +
  geom_point() +
  labs(
    title = "Scatter Plot of Corolla Diameter vs. Proportion Surviving to Flower",
    x = "SLA",
    y = "Proportion Surviving to Flower"
  ) +
  theme_minimal()
```

```
## Warning: Removed 283 rows containing missing values (`geom_point()`).
```



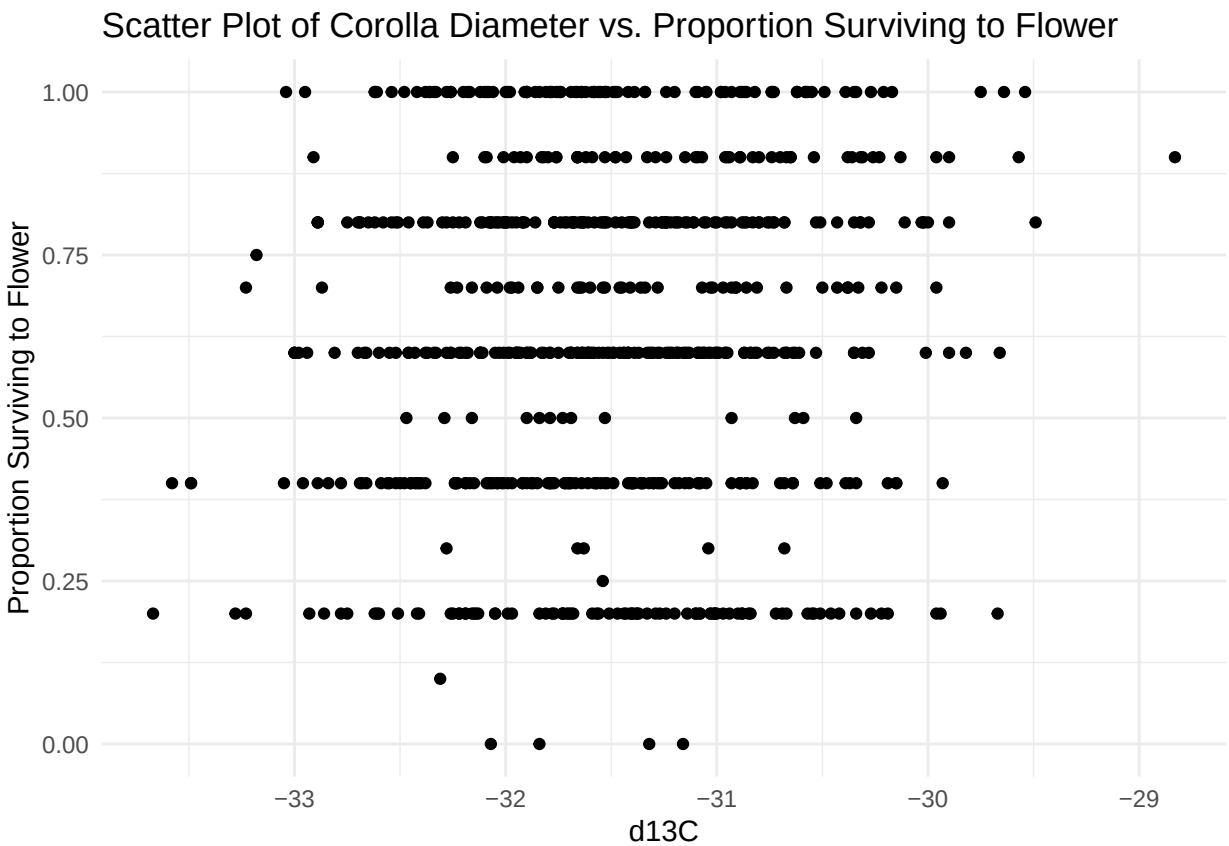

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = LMA_SEG, y = prop_surv_to_flower)) +
  geom_point() +
  labs(
    title = "Scatter Plot of Corolla Diameter vs. Proportion Surviving to Flower",
    x = "LMA",
    y = "Proportion Surviving to Flower"
  ) +
  theme_minimal()
```

```
## Warning: Removed 283 rows containing missing values (`geom_point()`).
```



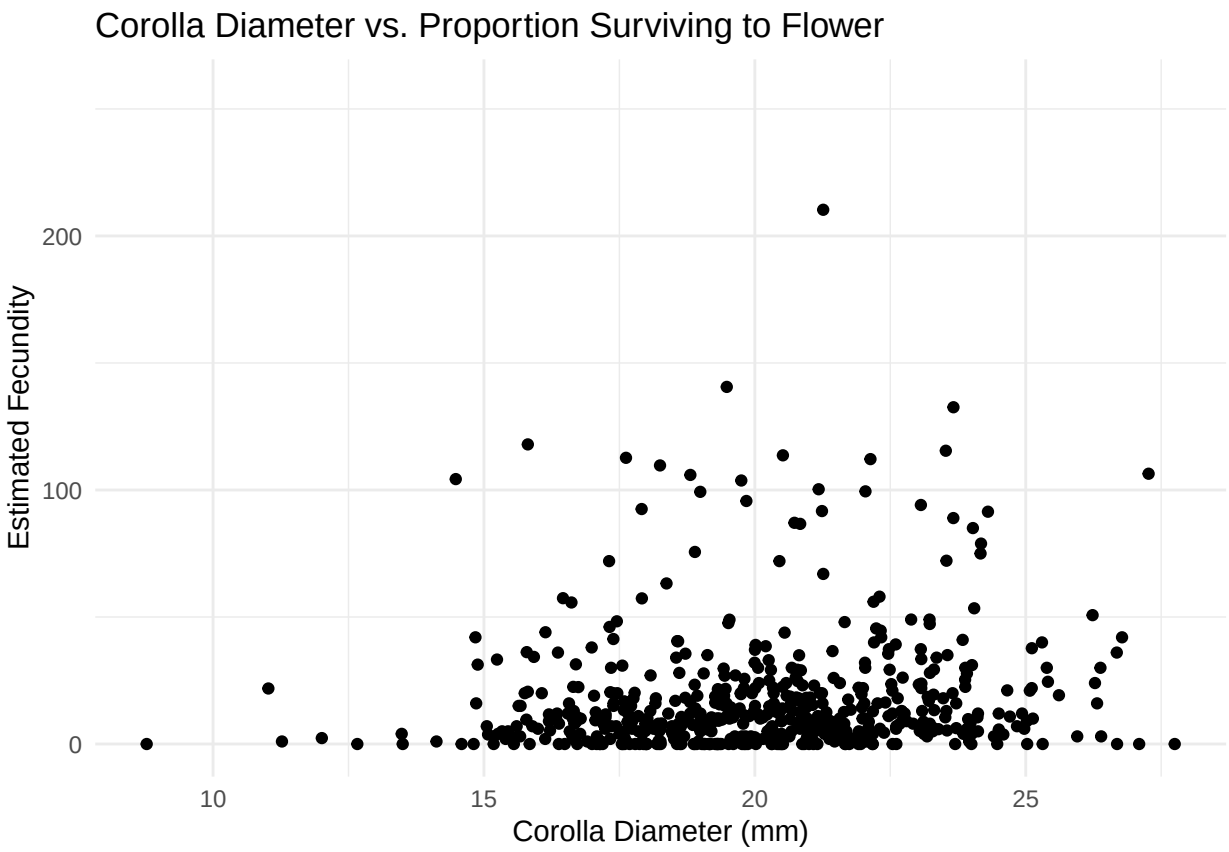

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = d13C_SEG, y = prop_surv_to_flower)) +
  geom_point() +
  labs(
    title = "Scatter Plot of Corolla Diameter vs. Proportion Surviving to Flower",
    x = "d13C",
    y = "Proportion Surviving to Flower"
  ) +
  theme_minimal()
```

```
## Warning: Removed 181 rows containing missing values (`geom_point()`).
```



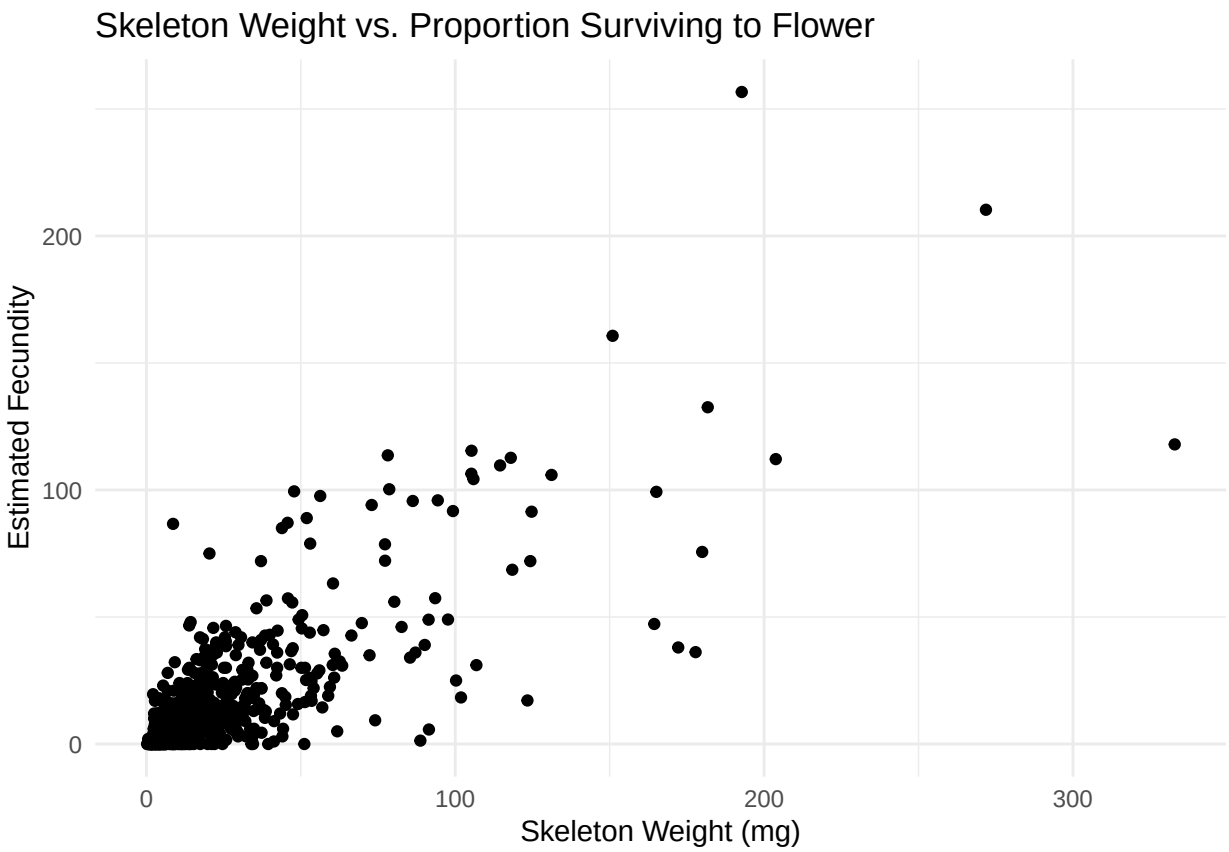

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = corolla_diam_mm_SEG, y = est_fecundity)) +
  geom_point() +
  labs(
    title = "Corolla Diameter vs. Proportion Surviving to Flower",
    x = "Corolla Diameter (mm)",
    y = "Estimated Fecundity"
  ) +
  theme_minimal()
```

```
## Warning: Removed 310 rows containing missing values (`geom_point()`).
```



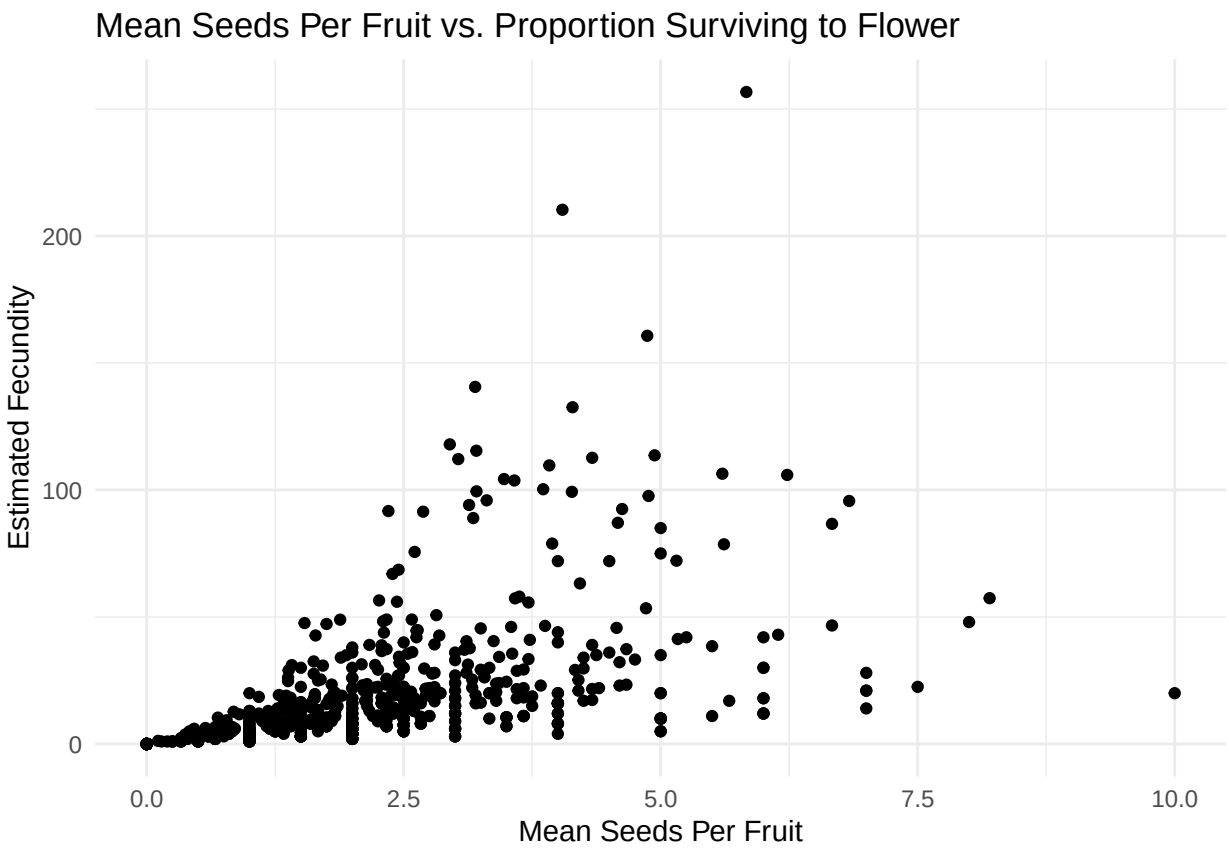

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = skel_dryweight_mg_SEG, y = est_fecundity)) +
  geom_point() +
  labs(
    title = "Skeleton Weight vs. Proportion Surviving to Flower",
    x = "Skeleton Weight (mg)",
    y = "Estimated Fecundity"
  ) +
  theme_minimal()
```

```
## Warning: Removed 236 rows containing missing values (`geom_point()`).
```



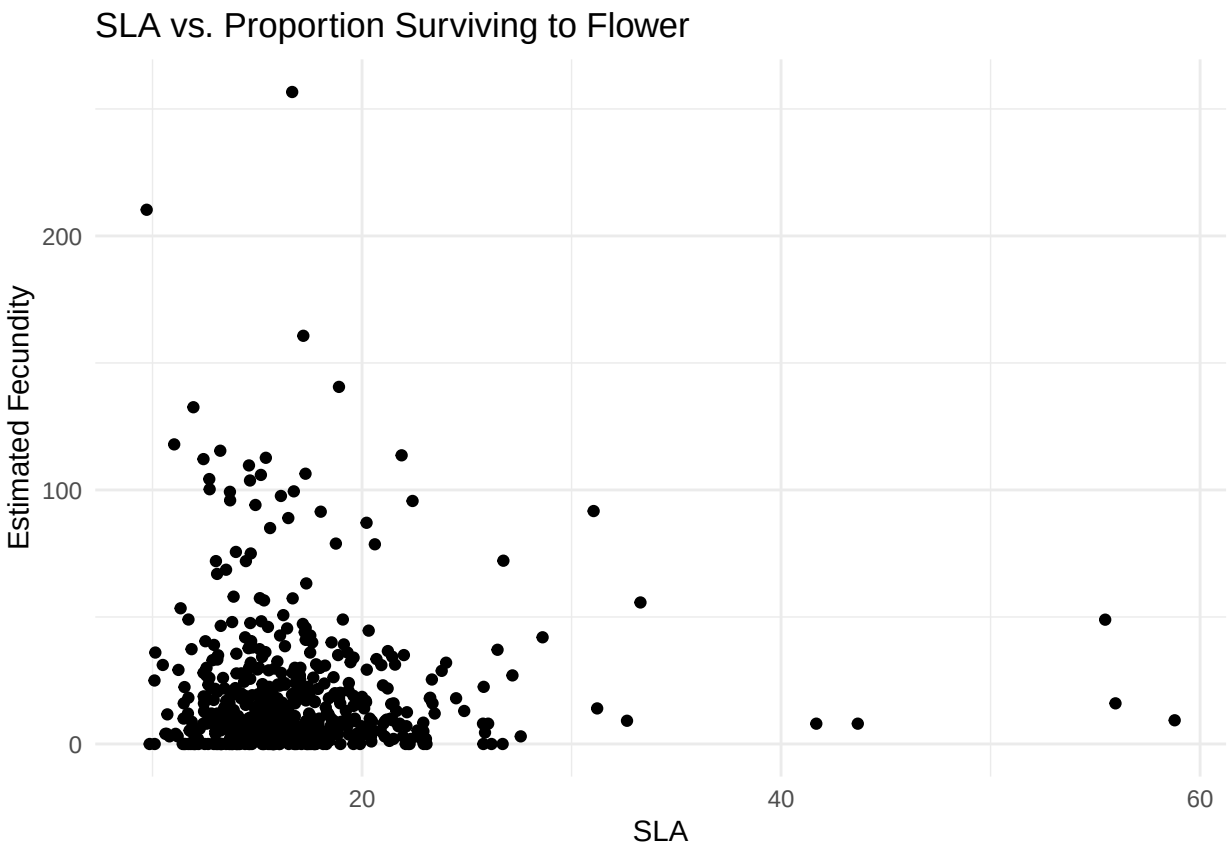

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = mean_seeds_per_fruit, y = est_fecundity)) +
  geom_point() +
  labs(
    title = "Mean Seeds Per Fruit vs. Proportion Surviving to Flower",
    x = "Mean Seeds Per Fruit",
    y = "Estimated Fecundity"
  ) +
  theme_minimal()
```

```
## Warning: Removed 150 rows containing missing values (`geom_point()`).
```



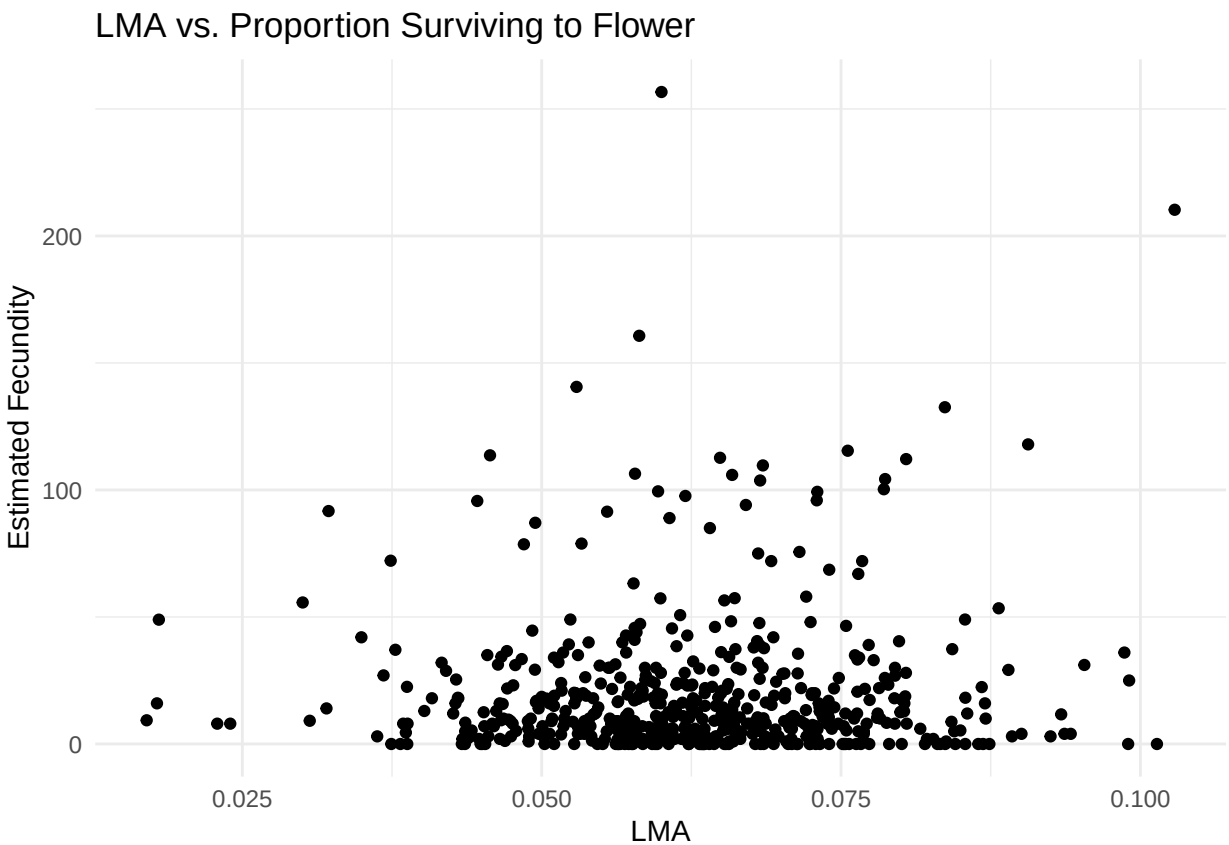

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = SLA_SEG, y = est_fecundity)) +
  geom_point() +
  labs(
    title = "SLA vs. Proportion Surviving to Flower",
    x = "SLA",
    y = "Estimated Fecundity"
  ) +
  theme_minimal()
```

```
## Warning: Removed 319 rows containing missing values (`geom_point()`).
```



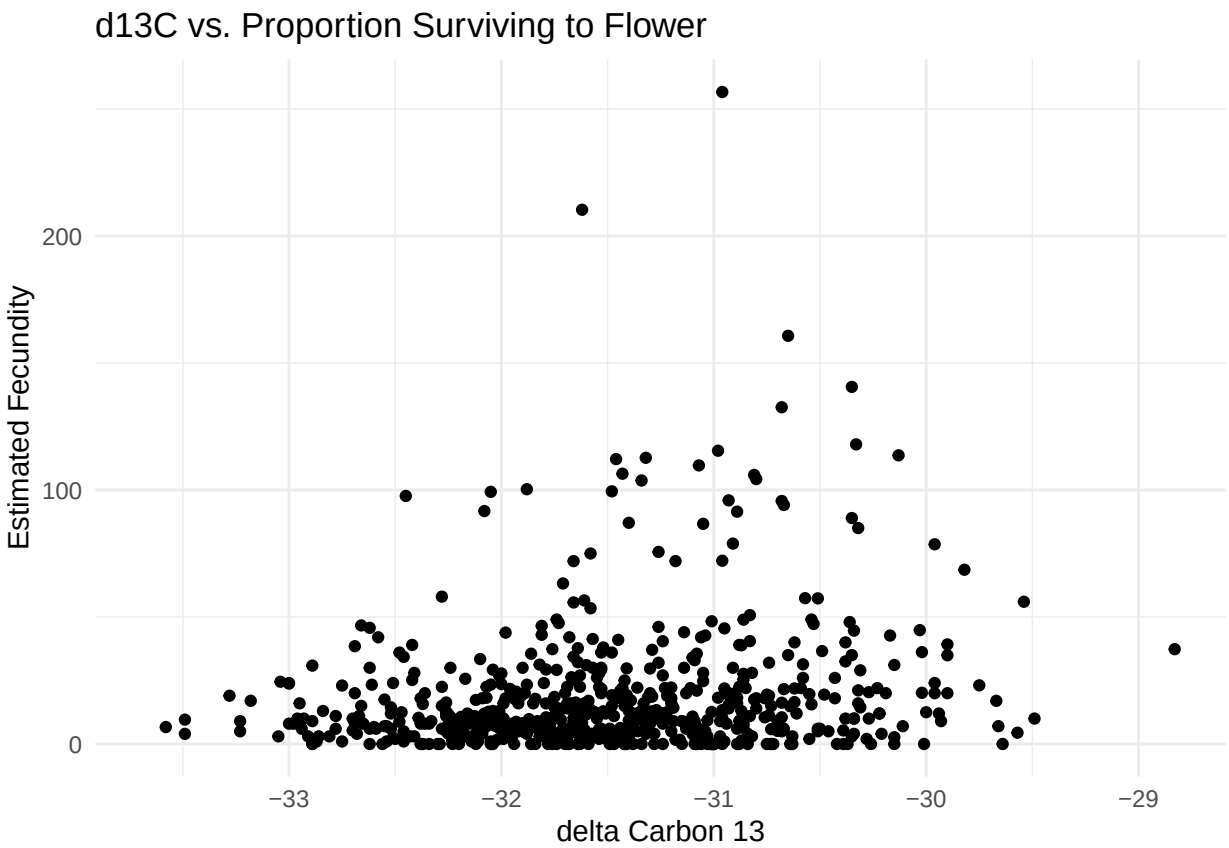

```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = LMA_SEG, y = est_fecundity)) +
  geom_point() +
  labs(
    title = "LMA vs. Proportion Surviving to Flower",
    x = "LMA",
    y = "Estimated Fecundity"
  ) +
  theme_minimal()
```

```
## Warning: Removed 319 rows containing missing values (`geom_point()`).
```




```r
# Create the data snooping scatterplot
ggplot(data = AC_22_23_fit, aes(x = d13C_SEG, y = est_fecundity)) +
  geom_point() +
  labs(
    title = "d13C vs. Proportion Surviving to Flower",
    x = "delta Carbon 13",
    y = "Estimated Fecundity"
  ) +
  theme_minimal()
```

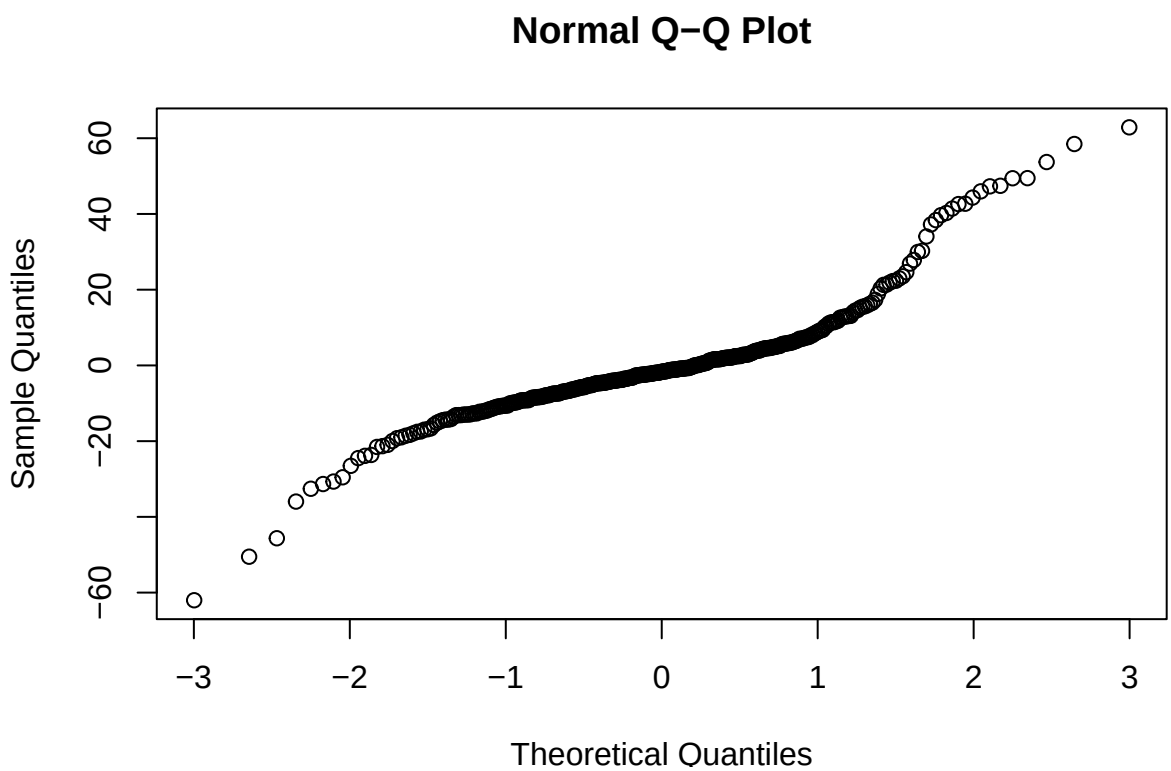
```
## Warning: Removed 237 rows containing missing values (`geom_point()`).
```



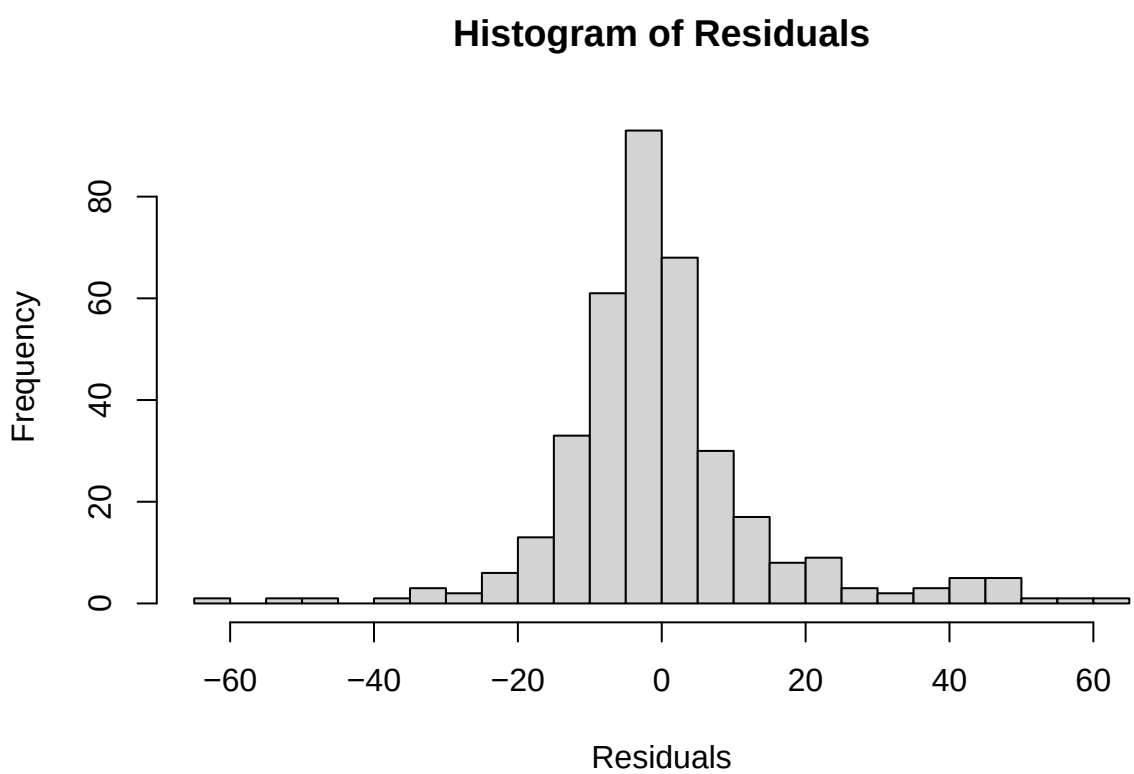

```
#corolla_daim and Skel dryweight model
mod_1 <- lmer(est_fecundity ~ corolla_diam_mm + skel_dryweight_mg + Transect + (1 | Donor)  + (1 | Dono

residuals <- resid(mod_1)

# Q-Q plot for normality
qqnorm(residuals) #good
```




```r
# Histogram for normality
hist(residuals, breaks = 20, main = "Histogram of Residuals", xlab = "Residuals") #normal
```




```r
AIC(mod_1)
```

```
## [1] 3112.336
```

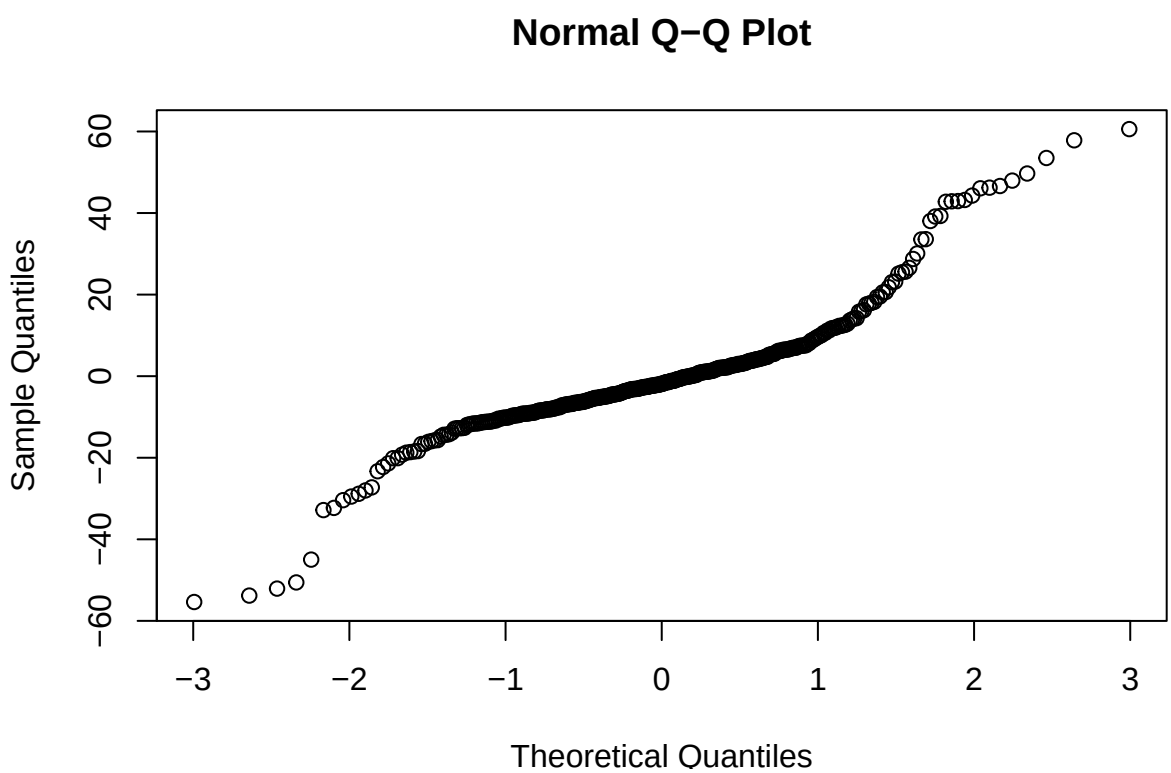
```r
mod_1
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: est_fecundity ~ corolla_diam_mm + skel_dryweight_mg + Transect +
##     (1 | Donor) + (1 | Donor:Recipient)
##    Data: AC_22_23_fit
## REML criterion at convergence: 3098.336
## Random effects:
##  Groups          Name        Std.Dev.
##  Donor:Recipient (Intercept)  4.746
##  Donor           (Intercept)  1.863
##  Residual                    15.568
## Number of obs: 368, groups:  Donor:Recipient, 105; Donor, 40
## Fixed Effects:
##       (Intercept)    corolla_diam_mm  skel_dryweight_mg           Transect
##            1.2057             0.4217             0.5462            -0.9374
```

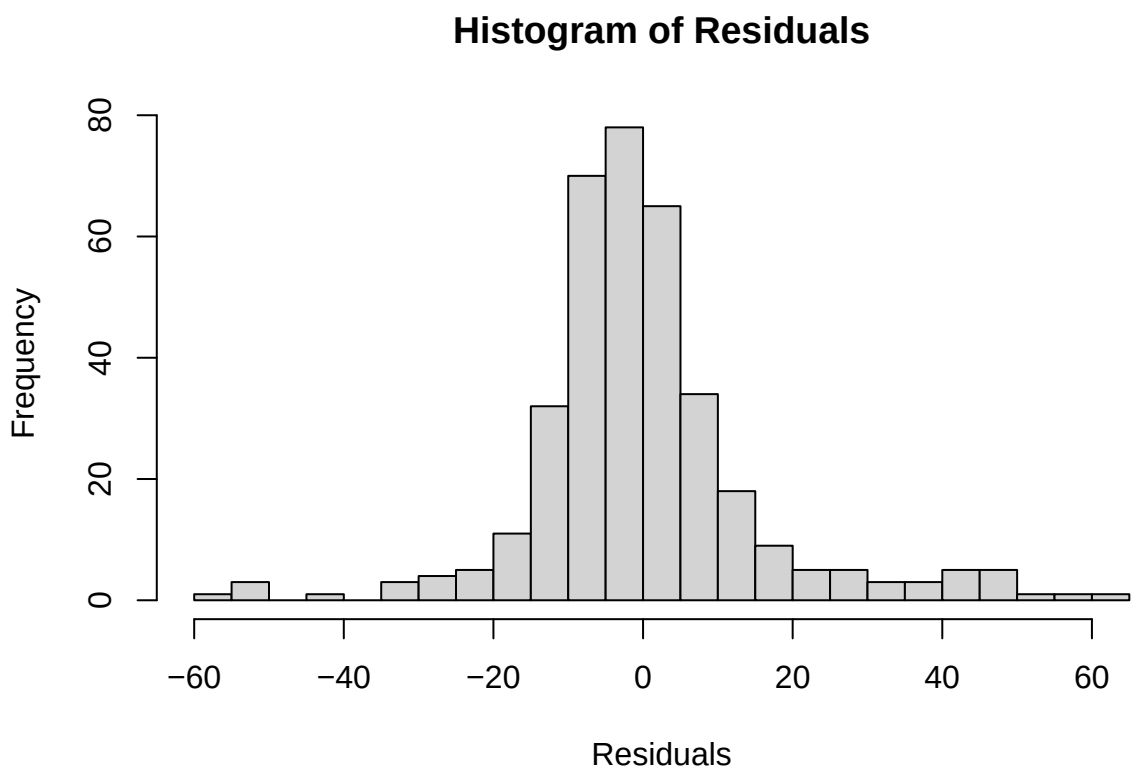
```r
# now include SLA
mod_2 <- lmer(est_fecundity ~ corolla_diam_mm_SEG + skel_dryweight_mg_SEG + SLA_SEG + Transect + (1 | D

residuals <- resid(mod_2)

# Q-Q plot for normality
qqnorm(residuals) #good
```




```r
# Histogram for normality
hist(residuals, breaks = 20, main = "Histogram of Residuals", xlab = "Residuals") #normal
```




```r
AIC(mod_2)
```

```
## [1] 3109.192
```

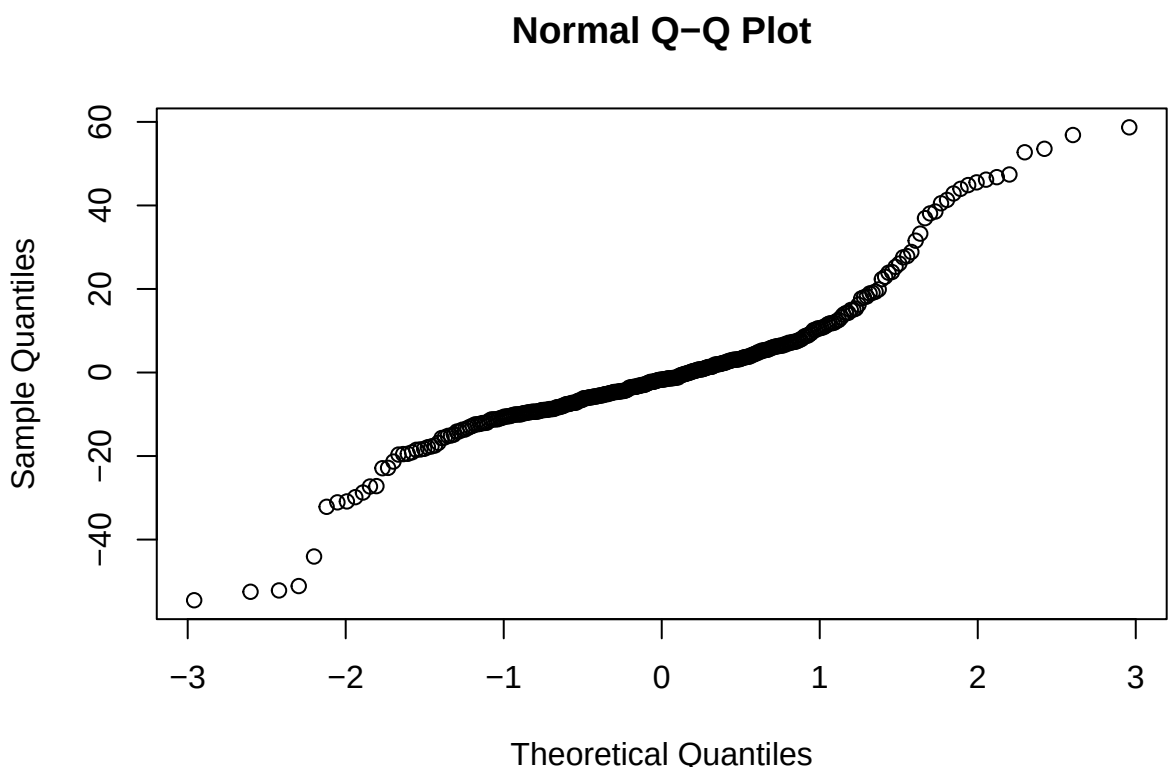
```r
mod_2
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: est_fecundity ~ corolla_diam_mm_SEG + skel_dryweight_mg_SEG +
##     SLA_SEG + Transect + (1 | Donor) + (1 | Donor:Recipient)
##    Data: AC_22_23_fit
## REML criterion at convergence: 3093.192
## Random effects:
##  Groups          Name        Std.Dev.
##  Donor:Recipient (Intercept)  5.311
##  Donor           (Intercept)  2.694
##  Residual                    16.221
## Number of obs: 363, groups:  Donor:Recipient, 106; Donor, 40
## Fixed Effects:
##           (Intercept)    corolla_diam_mm_SEG  skel_dryweight_mg_SEG
##               -7.4195                 0.6230                 0.5412
##               SLA_SEG               Transect
##                0.2472                -1.0480
```

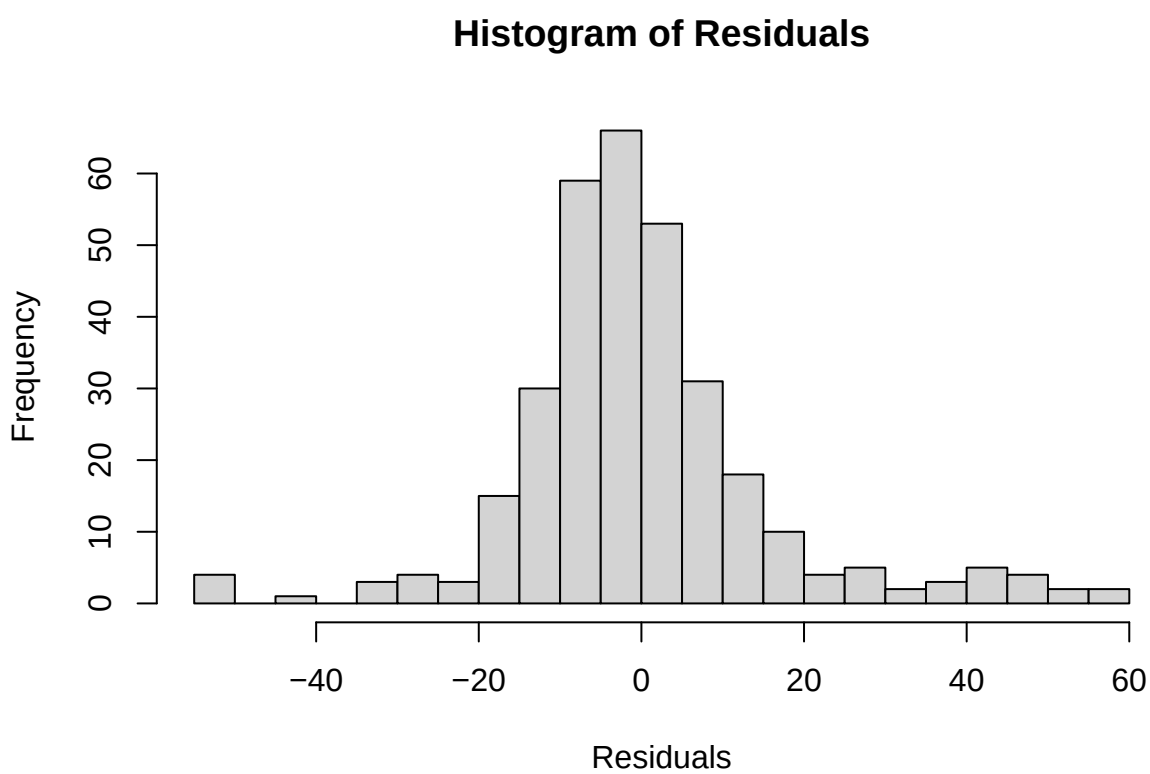
```r
# now include d13C
mod_3 <- lmer(est_fecundity ~ corolla_diam_mm_SEG + skel_dryweight_mg_SEG + SLA_SEG + d13C + Transect +

residuals <- resid(mod_3)

# Q-Q plot for normality
qqnorm(residuals) #good
```




```r
# Histogram for normality
hist(residuals, breaks = 20, main = "Histogram of Residuals", xlab = "Residuals") #normal
```




```r
AIC(mod_3)
```

```
## [1] 2804.022
```

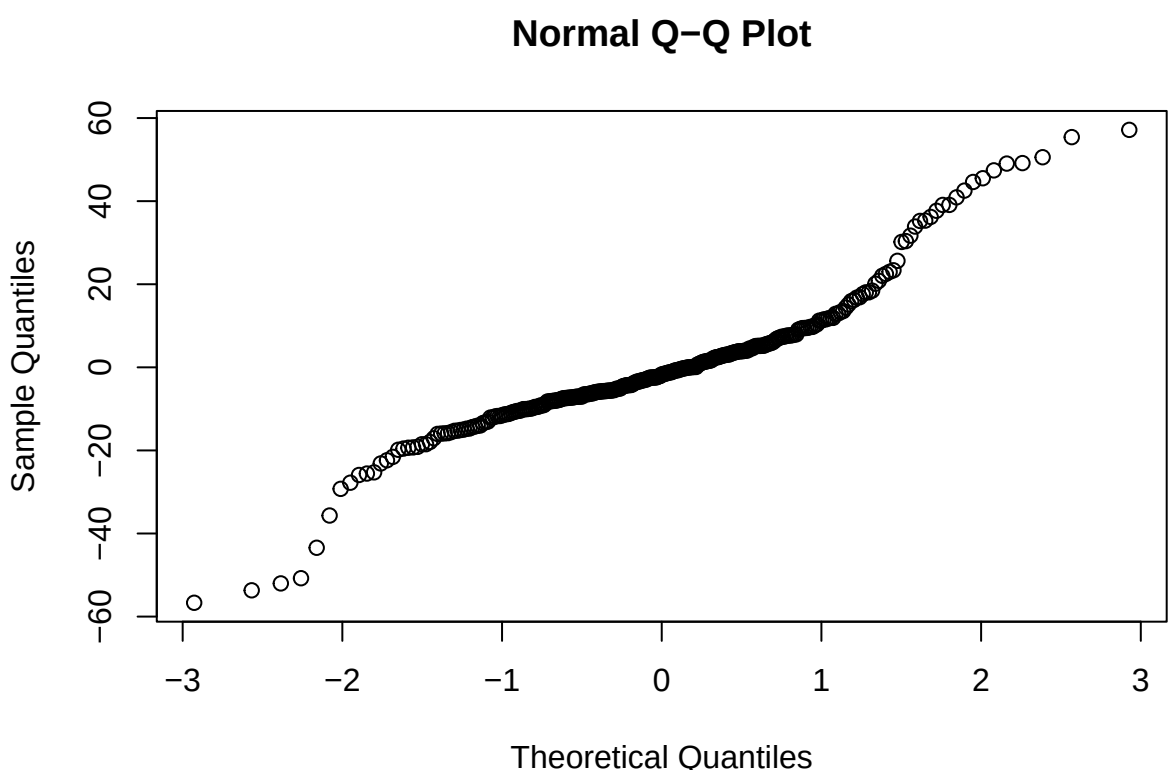
```r
mod_3
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: est_fecundity ~ corolla_diam_mm_SEG + skel_dryweight_mg_SEG +
##     SLA_SEG + d13C + Transect + (1 | Donor) + (1 | Donor:Recipient)
##    Data: AC_22_23_fit
## REML criterion at convergence: 2786.022
## Random effects:
##  Groups          Name        Std.Dev.
##  Donor:Recipient (Intercept)  5.375
##  Donor           (Intercept)  2.990
##  Residual                    16.993
## Number of obs: 324, groups:  Donor:Recipient, 106; Donor, 40
## Fixed Effects:
##           (Intercept)    corolla_diam_mm_SEG  skel_dryweight_mg_SEG
##               61.7599                 0.6167                 0.5269
##               SLA_SEG                   d13C               Transect
##                0.2679                 2.1579                -1.2388
```

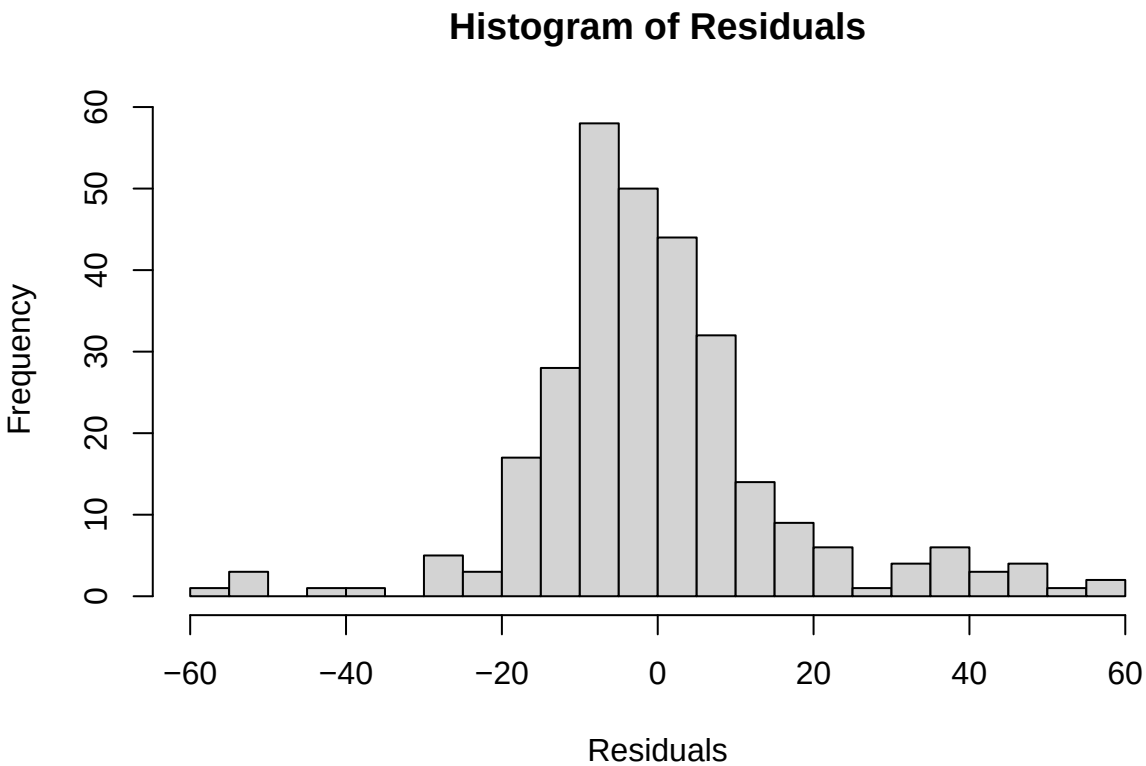
```r
# now include SLA
mod_4 <- lmer(est_fecundity ~ corolla_diam_mm_SEG + skel_dryweight_mg_SEG + SLA_SEG + d13C + msm_all + 

residuals <- resid(mod_4)

# Q-Q plot for normality
qqnorm(residuals) #good
```




```r
# Histogram for normality
hist(residuals, breaks = 20, main = "Histogram of Residuals", xlab = "Residuals") #normal
```




```r
AIC(mod_4)
```

```
## [1] 2528.428
```

```r
mod_4
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: est_fecundity ~ corolla_diam_mm_SEG + skel_dryweight_mg_SEG +
##     SLA_SEG + d13C + msm_all + Transect + (1 | Donor) + (1 |
##     Donor:Recipient)
##    Data: AC_22_23_fit
## REML criterion at convergence: 2508.428
## Random effects:
##  Groups          Name        Std.Dev.
##  Donor:Recipient (Intercept)  4.007
##  Donor           (Intercept)  2.818
##  Residual                    17.106
## Number of obs: 293, groups:  Donor:Recipient, 104; Donor, 40
## Fixed Effects:
##           (Intercept)    corolla_diam_mm_SEG  skel_dryweight_mg_SEG
##              104.7797                 0.5596                 0.5211
##               SLA_SEG                  d13C               msm_all
##                0.1928                 2.9575               -7.5691
##              Transect
##               -1.2860
```